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Online Quiz Application

A Scalable Web Architecture Built with Core Java & Cloud Persistence

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Executive Introduction

Welcome to our Online Quiz Application. This project serves as a robust digital alternative to legacy paper examinations, ensuring transparency and lightning-fast evaluation.

By migrating tests to a cloud-powered environment, we eliminate logistical hurdles and human errors in scoring.



Project Objective

To build a high-performance, platform-independent assessment engine that provides real-time result analytics for both educators and students.

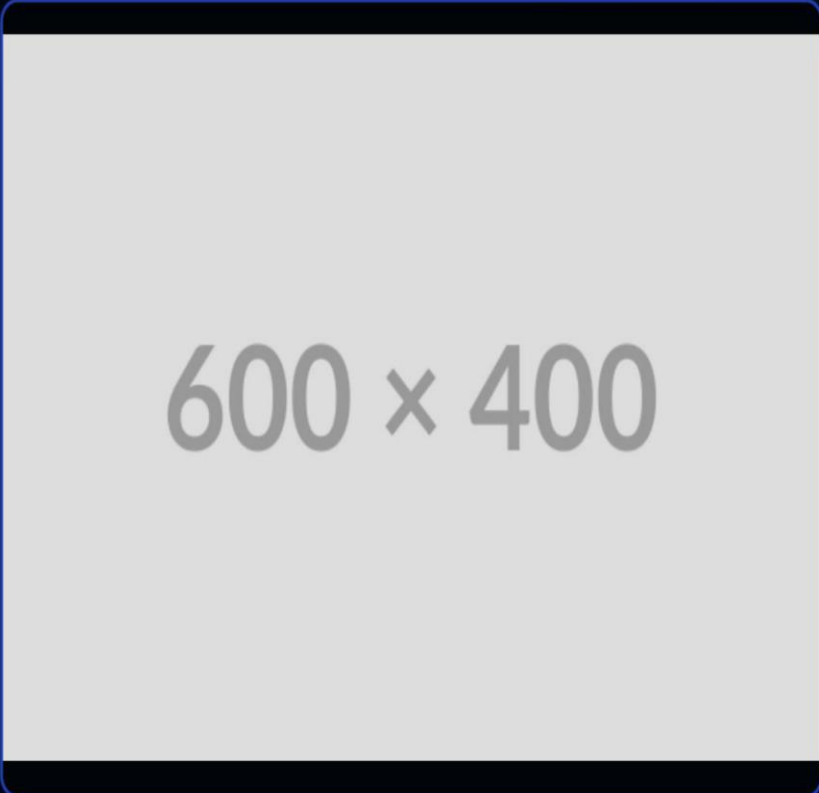
Working: Unified Gateway Logic

One Single Entry Page

To optimize the user experience, we designed a single login portal for both Teachers and Students. The system identifies the user type at the backend immediately upon credential submission.

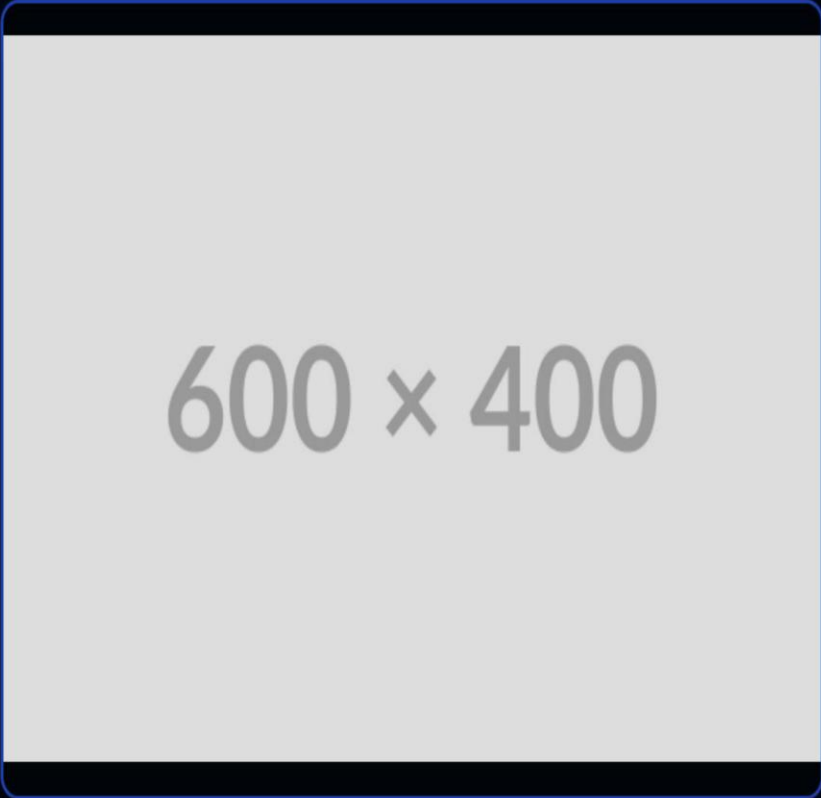
OOP Concept: Encapsulation

We encapsulate user data within secure objects, ensuring that credentials are never exposed during transmission or routing.



600 × 400

Working: Intelligent Routing



600 × 400

Dynamic Redirection Flow

Once the system validates the input, the application performs an intelligent redirection based on the detected profile:

- **Teachers:** Automatically routed to the Management Hub for quiz creation.
- **Students:** Automatically routed to the live Exam Engine via a secure Test Code.

OOP Concept: Polymorphism

The same entry controller executes different routing behaviors depending on the identity detected during runtime.

Working: Automated Feedback



Instant Scoring

The moments a student submits their quiz, the backend engine calculates the result instantly and renders it on the student's dashboard.



Automated Emailing

Upon exam expiry, the system triggers an automated email dispatch, sending the final rank and scorecard to the student's registered address.



Reliable Persistence

Using Firebase, we ensure scores are saved permanently in real-time, even if the user experiences network fluctuations during submission.

Technical Foundations & Mapping

Mapping to Java Curriculum

- **Memory Mastery:** Efficient Stack/Heap management during live question fetching.
- **Defensive Programming:** Robust try-catch blocks to handle cloud connectivity issues.
- **Abstraction:** Hiding complex NoSQL database queries behind simple User Interfaces.

Project main code

```
@WebServlet("/StartExamEngine")
public class StudentExamController extends HttpServlet {
    private final Firestore db = FirebaseRegistry.getDatabase();

    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException {
        try {
            String email = request.getParameter("studentEmail");
            String testCode = request.getParameter("testCode");

            if (testCode != null) {
                testCode = testCode.trim();
            }

            // Firebase se 'exams' collection ke andar test document fetch karna
            DocumentSnapshot examDoc = db.collection("exams").document(testCode).get().get();

            if (examDoc.exists()) {
                HttpSession session = request.getSession();
                session.setAttribute("studentEmail", email);

                Map<String, Object> examData = examDoc.getData();
                request.setAttribute("examData", examData);

                // Pure dynamic data ke sath student ko test room me forward karna
                request.getRequestDispatcher("/student_exam_room.jsp").forward(request, response);
            } else {
                response.sendRedirect("student_portal.jsp?error=CodeInvalid");
            }
        }
    }
}
```

Project outlook

Welcome to Exam Portal

Enter credentials to log in as Teacher or Student

Email Address

teacher@gmail.com

Password / Test Authentication Code

Login / Start Test



Test Submitted!

Aapka response safely cloud par save ho gaya hai.

1 Points



Total Time Taken: 0 Mins 10 Secs

Note: System caught tab switching behavior. Test was auto-submitted.



Note: Aapki final comparative rank test window close hone ke baad aapke mail pe bhej di jayegi.

Future Scope & Evolutionary Horizons

Upcoming Milestones

- **AI Proctoring:** Real-time webcam monitoring to detect cheating behaviors.
- **Adaptive Learning:** Quizzes that change difficulty based on student performance.
- **Microservices:** Decoupling features for horizontal global scaling.



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Conclusion

"Our Online Quiz Application successfully implements modern Object-Oriented paradigms to build a secure, fast, and automated educational ecosystem. It proves that Java and Cloud technologies can revolutionize legacy assessment workflows."

THANK YOU FOR YOUR ATTENTION!
